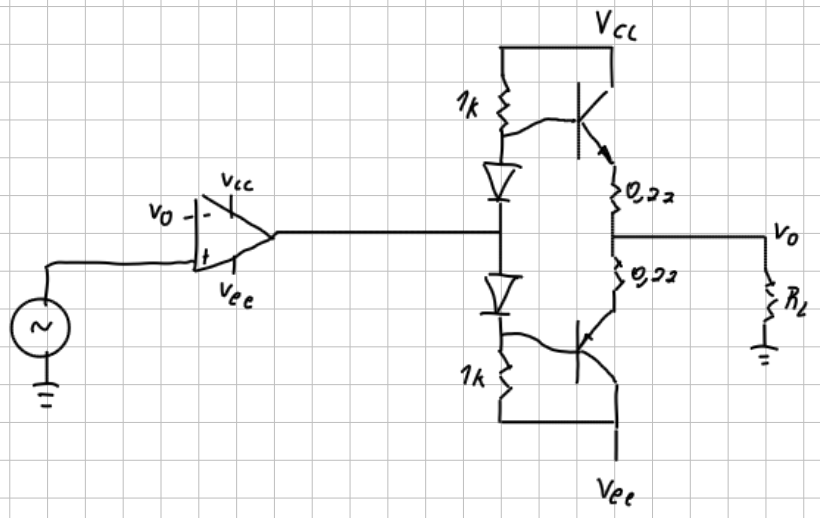
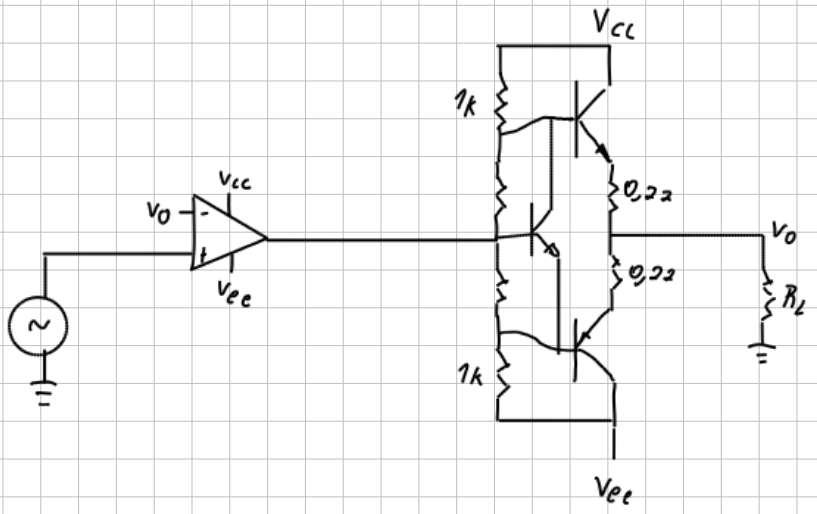
Amplificador Clase B





Multiplicador Vbe

$$I_{R7} = \frac{V_{be}}{R_7}$$

$$V_{R6} = R_6 (I_{R7} + I_b)$$

$$V_{R6} = R_6 \frac{V_{be}}{R_7} + R_6 I_B$$

$$V_{CE} = V_{BE} + R_6 \frac{V_{be}}{R_7} + R_6 I_B$$

$$V_{CE} = V_{BE}\left(1 + \frac{R_6}{R_7}\right) + \overbrace{R_6 I_B}^{\approx 0}$$

$V_{Darlinton} = 2{,}8\ V$

Polarizacion 50 - 100 mA

$$P_L = \frac{V_{LRMS}^2}{R_L} = \frac{V_{LPP}^2}{2R_L}$$

$$P_S = 2V_{CC}\frac{I_{LP}}{\pi} \qquad I_{LP} = \frac{V_{LP}}{R_L}$$

$$\eta = \frac{P_L}{P_S}$$

$$\eta = \frac{V_{LP}^2/2R_L}{\frac{2V_{CC}\ V_{LP}}{\pi}}$$

$$\boxed{\eta = \frac{V_{LP}\ \pi}{4\ V_{CC}}}$$

$$\eta_{max} = \frac{\pi}{4} \approx 78{,}5\ \% \qquad V_{LP} = V_{CC}$$

$$P_D = P_S - P_L$$

$$P_D = \frac{2V_{CC}V_{LP}}{\pi R_L} - \frac{V_{LP}^2}{2R_L}$$

$$P_{Dmax}\Big|_{V_{LP}=\frac{2V_{CC}}{\pi}} = \frac{4V_{CC}^2}{\pi R_L} - \frac{4V_{CC}^2}{2\pi^2 R_L}$$

$$\boxed{P_{Dmax} = \frac{V_{CC}^2}{\pi^2 R_L}}$$

2 Transistores

$V_{CE} = 0,7\left(1 + 1\right)$

$V_{CE} \approx 1,4\,V$

$P_{Omax} = \frac{15\,V^2}{\pi^2 \cdot 8\,\Omega}$

$P_{Omax} = 2,85\,W$ por mosFet

$P_{Imax} = \frac{V_{pp}^2}{2 R_L}$

$P_{Imax} = \frac{12^2}{2 \cdot 8\,\Omega}$ → ±12V empieza a saturar

$P_{Imax} \approx 9\,W$

$P_{Smax} = \frac{2 V_{CC} V_{cp}}{\pi R_L}$

$P_{Smax} = 14,32\,W$